INPUT, STRIP() & CONVERSION

`input()` returns text; `.strip()` trims the edges; convert before you calculate.

1) Read a name and clean it

Example :

```
name = input("Enter a name: ").strip()
print("Welcome,", name)
```

Output :

```
Enter a name:    Asha
Welcome, Asha
```

Method	What it trims	Result on " Python class "
`strip()`	both ends	`'Python class'`
`lstrip()`	left side only	`'Python class  '`
`rstrip()`	right side only	`'  Python class'`

2) Type mismatch

Example :

```
number = input("Enter your number: ")
print(f"Hello {number}", type(number))
print(number + number)
```

Output :

```
Enter your number: 25
Hello 25 <class 'str'>
2525
```

typing digits does not make a number: "25" + "25" joins the text into "2525"

3) Convert with int() and float()

Example :

```
price = float(input("Enter item price: "))
quantity = int(input("Enter quantity: "))
total = price * quantity
print("Total:", total)
```

Output :

```
Enter item price: 49.5
Enter quantity: 3
Total: 148.5
```

Call	What it does	Result
`int("25")`	text to whole number	25
`float("25.5")`	text to decimal	25.5
`int("25.5")`	not an integer literal	`ValueError`

Safe Input & F-strings

Handle bad input with `try / except`, then shape output with `print()` and f-strings.

1) Handle invalid numbers

Example :

```
age_text = input("Enter your age: ").strip()
try:
    age = int(age_text)
except ValueError:
    print("Please enter a whole number, such as 18.")
else:
    print(f"Next year you will be {age + 1}.")
```

Output :

```
Enter your age: abc
Please enter a whole number, such as 18.
```

try runs the risky line; except runs only on a ValueError; else runs when no error

2) print(): sep and end

Example :

```
print("Python", "Basics", "Day 2")
print("2026", "10", "13", sep="-")
print("Loading", end="...")
print("Done!")
```

Output :

```
Python Basics Day 2
2026-10-13
Loading...Done!
```

3) f-strings and number formats

Example :

```
name = "Asha"
age = 19
marks = 87.456
print(f"{name} is {age} years old.")
print(f"Marks: {marks:.2f}")
print(f"{name} scored {marks:.1f} marks.")
print(f"{0.875:.1%}")
```

Output :

```
Asha is 19 years old.
Marks: 87.46
Asha scored 87.5 marks.
87.5%
```

Format	Meaning	Result
`{marks:.2f}`	2 decimal places (rounded)	87.46
`{marks:.1f}`	1 decimal place	87.5
`{pct:.1%}`	multiply by 100, add %	87.5%

formatting changes only how a value is shown, marks still holds 87.456

String Basics & Project

Strings are **immutable**: methods return a new string, so save the result.

1) Join, repeat, count, index

Example :

```
first_name = "Asha"
last_name = "Patil"
full_name = first_name + " " + last_name
print(full_name)
print("Hi! " * 3)
print("Length:", len(full_name))
print(full_name[0], full_name[-1])
```

Output :

```
Asha Patil
Hi! Hi! Hi!
Length: 10
A l
```

indexing starts at 0; the last character is always -1

2) Handy string methods

Method	Example	Result
`.upper()` / `.lower()`	`"Hi".upper()`	`'HI'`
`.title()`	`"asha patil".title()`	`'Asha Patil'`
`.split(",")`	`"a,b,c".split(",")`	`['a', 'b', 'c']`

3) Mini project: student marks report

Example :

```
name = input("Student name: ").strip()
marks = float(input("Marks obtained: "))
total_marks = float(input("Total marks: "))
percentage = marks / total_marks
print("--- Student Report ---")
print(f"Name : {name}")
print(f"Marks : {marks:.1f} / {total_marks:.1f}")
print(f"Percentage : {percentage:.1%}")
```

Output :

```
Student name:   Asha
Marks obtained: 87.5
Total marks: 100
--- Student Report ---
Name : Asha
Marks : 87.5 / 100.0
Percentage : 87.5%
```

strip cleaned the spaces around the name, the formats shaped the numbers

Quick check

`"Age: " + 19` raises a **TypeError**: fix it with `str(19)` or an f-string.

Errors & Practice

Know the common errors, then try each task before comparing with the solution.

1) Common errors

Error	When it happens	Example
NameError	name is not defined	print(scroe)
ValueError	text cannot become a number	int("hello")
TypeError	mixing text and a number	"Age: " + 19
Case matters	input() is not Input()	Input("Name: ")

fix the TypeError with str(19) or an f-string

2) Practice tasks

Task 1 Name and city greeting.

Solution :

```
name = input("Name: ").strip()
city = input("City: ").strip()
print(f"Hello {name}, welcome to {city}!")
```

```
Name: Ravi
City: Pune
Hello Ravi, welcome to Pune!
```

Task 2 Price x quantity.

Solution :

```
price = float(input("Price: "))
quantity = int(input("Quantity: "))
print(f"Total: {price * quantity:.2f}")
```

```
Price: 49.5
Quantity: 3
Total: 148.50
```

Task 3 Decimal as percentage.

Solution :

```
x = float(input("Decimal 0 to 1: "))
print(f"{x:.1%}")
```

```
Decimal 0 to 1: 0.875
87.5%
```

Task 4 Full name in title case.

Solution :

```
first = input("First name: ").strip()
last = input("Last name: ").strip()
print(f"{first} {last}".title())
```

```
First name: asha
Last name: patil
Asha Patil
```

Task 5 Length, UPPER, lower.

Solution :

```
text = input("Sentence: ")
print(len(text), text.upper(), text.lower())
```

```
Sentence: Python is fun
13 PYTHON IS FUN python is fun
```

Task 6 Split three foods.

Solution :

```
foods = input("Three foods: ").split(",")
print(foods)
```

```
Three foods: pizza,pasta,idli
['pizza', 'pasta', 'idli']
```

Quick check

`input()` always returns **str**; `f"{marks:.2f}"` with 87.456 shows **87.46**; `len("Asha Patil")` is **10**.